WORKSHEET

Approximation with derivatives

1 Given $f(x) = x^3$, find:

a the derivative of $f(x)$

b the value of $f(x)$ at $x = 12$

c the slope of $f(x)$ at $x = 12$

d an approximation for $f(12.02)$ using the slope of the function at $x = 12$.

2 Find an approximation for 15.95^3.

3 Find an approximation for $\sqrt{101}$ by differentiating $f(x) = \sqrt{x}$.

4 Find an approximate value of:

a $\sqrt{24}$

b $\sqrt{81.01}$

c $\sqrt{28}$

d $\sqrt{146}$

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5 Use derivatives to find an approximate value for 20.05^2.

6 Given $g(x) = 2x^3$, find:

a $g(8)$

b an approximation for $g(8.1)$

c the error between the approximation and the exact value of $g(8.1)$

d the error as a percentage of the exact answer

e the percentage error for the approximation of $g(8.4)$ using the derivative of g at $x = 8$

f the percentage error for the approximation of $g(8.9)$ using the derivative of g at $x = 8$.

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7 The radius of a circle is measured to within 1% error. What is the approximate percentage error of the area of a circle with radius 5 cm?

8 A side of a cube is measured with 2% error. Find:

a the approximate percentage error in the surface area of the cube

b the approximate percentage error in the volume of the cube.

9 The radius of a sphere is measured with 0.5% error. Find:

a the approximate percentage error in the surface area of the sphere

b the approximate percentage error in the volume of the sphere.

10 The volume of a cylinder is to be calculated using the measured length of its radius. If it is necessary for the calculated volume to have, at most, 2% error, what is the approximate maximum percentage error of the radius measurement?

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11 The equation of a function is given by $y = 4x^4 - 7x^3 + 2x - 1$. Find:

a the derivative of the function

b the slope of the function at $x = 2$

c the approximate change in y as x increases from $x = 2$ to $x = 2.02$.

12 The cost ($) of producing n soccer balls is given by $C(n) = 30 - 0.5n + 40\sqrt{n}$. Use derivatives to approximate the change in costs if orders are increased from 200 to 210.

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Answers

1 **a** $3x^2$

b 1728

c 432

d ≈ 1736.64

2 ≈ 4057.6

3 ≈ 10.05

4 **a** ≈ 4.9

b ≈ 9.00056

c ≈ 5.3

d ≈ 12.083

5 ≈ 402

6 **a** 1024

b ≈ 1062.4

c 0.482

d $\approx 0.045\%$

e $\approx 0.659\%$

f $\approx 2.861\%$

7 2%

8 **a** 4%

b 6%

9 **a** 1%

b 1.5%

10 1% (assuming the height of the cylinder is known exactly)

11 **a** $\frac{dy}{dx} = 16x^3 - 21x^2 + 2$

b 46

c ≈ 0.92

12 \$9.14

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